Danmeng Hu

Fashion Corporate Bonds

Financial Economics
29 August 2025 to 31 August 2026

Contents

Q1. Bond Selection and Prices

Q2. Yield to Maturity

Q3. Interpretation of Price and Yield Changes

Q4. Bond Pricing Calculation

Q5. Comparison Across Bonds

Appendix. Code, AI Use and Sources

Project Overview

- I compare the prices and yields of five fashion-related corporate bonds.
- The issuers are **Nike, Ralph Lauren, Tapestry, PVH and VF Corporation.**
    - All five bonds have fixed coupons, semiannual payments and USD principal.
- The sample runs from 29 August 2025 to 31 August 2026, with 252 observations per bond.
- I use these data to study the price-yield relationship, check a bond price, and compare required returns.

Q1(a). Selected Bonds (Table 1)

Company	CUSIP	Coupon (%)	Maturity	Face value ($)
Nike	654106AK9	2.85	2030-03-27	1,000
Ralph Lauren	751212AD3	5.00	2032-06-15	1,000
Tapestry	876030AA5	3.05	2032-03-15	1,000
PVH	693656AE0	5.50	2030-06-13	1,000
VF Corporation	918204AR9	6.00	2033-10-15	1,000

- All five bonds pay fixed coupons twice a year and are USD senior unsecured notes.

Q1(a). Bond Prices (Table 2)

- Prices are % of face value. The table only shows the first and last four dates (full version contains 252 dates and corresponding data points)

Date	NKE	RL	TPR	PVH	VFC
2025-08-29	94.930	102.058	89.550	101.950	94.876
2025-09-02	94.730	101.684	89.331	101.678	94.873
2025-09-03	94.908	102.163	89.626	101.918	95.036
2025-09-04	95.149	102.587	89.984	102.149	95.279
2026-08-26	93.903	100.145	90.032	100.503	97.288
2026-08-27	93.905	100.219	89.891	100.454	97.352
2026-08-28	93.644	99.752	89.513	100.179	97.580
2026-08-31	93.603	99.713	89.273	100.112	97.270

Q1(b). Company Selection (Table 3)

- I chose these companies because I follow their brands and am interested in how changes in consumer spending affect their sales and cash flow. Comparing their bonds lets me look beyond brand popularity and consider how investors assess each company's ability to meet its debt payments.
- The group includes sportswear, footwear and apparel companies with different customers and business models.
- I selected five issuers with usable bond histories for the bond analysis.

Bond issuers	Other companies in the tracking group
Nike (NKE)	lululemon (LULU)
Ralph Lauren (RL)	Deckers Outdoor (DECK)
Tapestry (TPR)	Levi Strauss (LEVI)
PVH (PVH)	Abercrombie & Fitch (ANF)
VF Corporation (VFC)	American Eagle Outfitters (AEO)

Q1(c). Bond Price Conversion

Nike bond example

- Bond: Nike (CUSIP 654106AK9)
- Date: 31 August 2026
- FactSet quote: 93.603% of face value
- Verified face value: $1,000

Calculation

Dollar price = (quote / 100) × face value

= (93.603 / 100) × $1,000

= 0.93603 × $1,000

= $936.03 per $1,000 face value

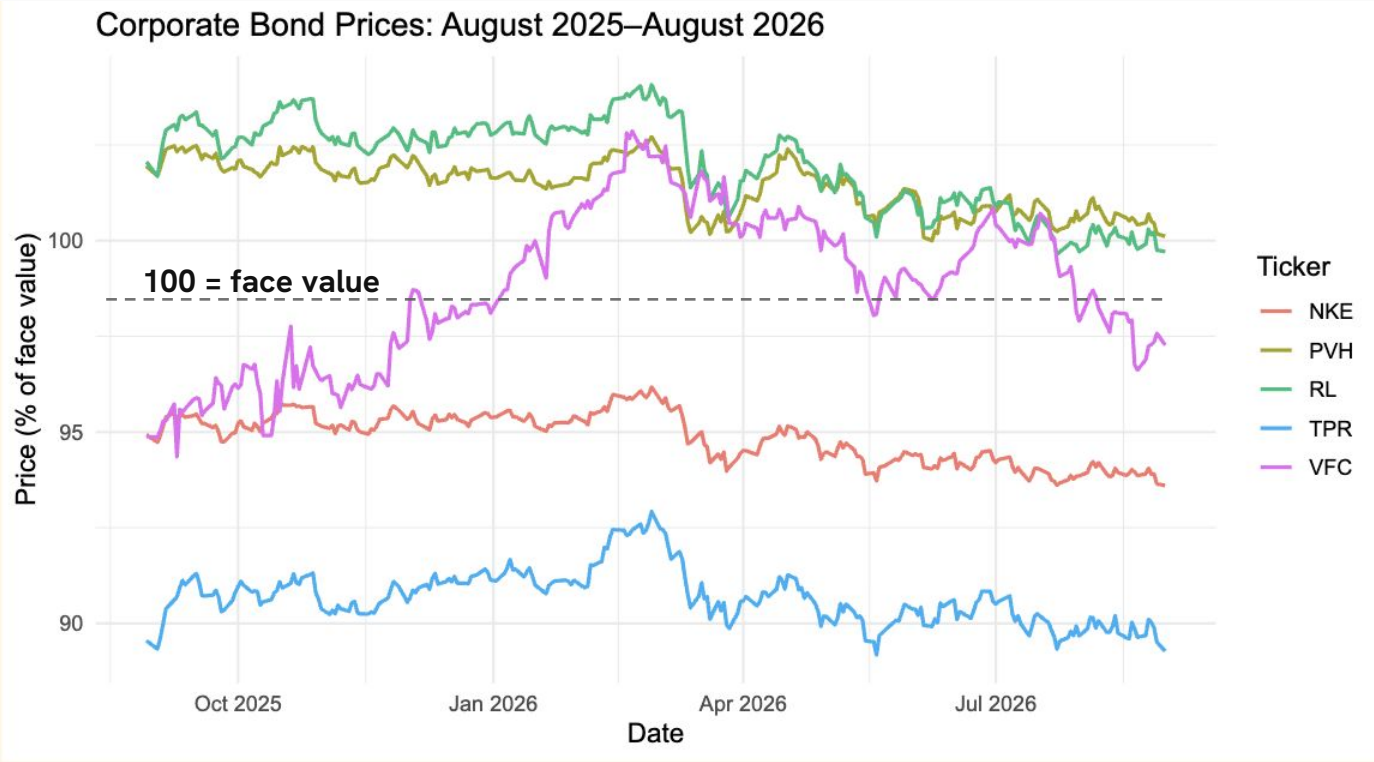
Face Value Verification

NIKE, Inc. · CUSIP 654106AK9

Face Value	1,000.00
Minimum Denomination	2,000.00
Minimum Increment	1,000.00

FactSet confirms a $1,000 face value

Q1(d). Bond Prices (Figure 1)



Price levels

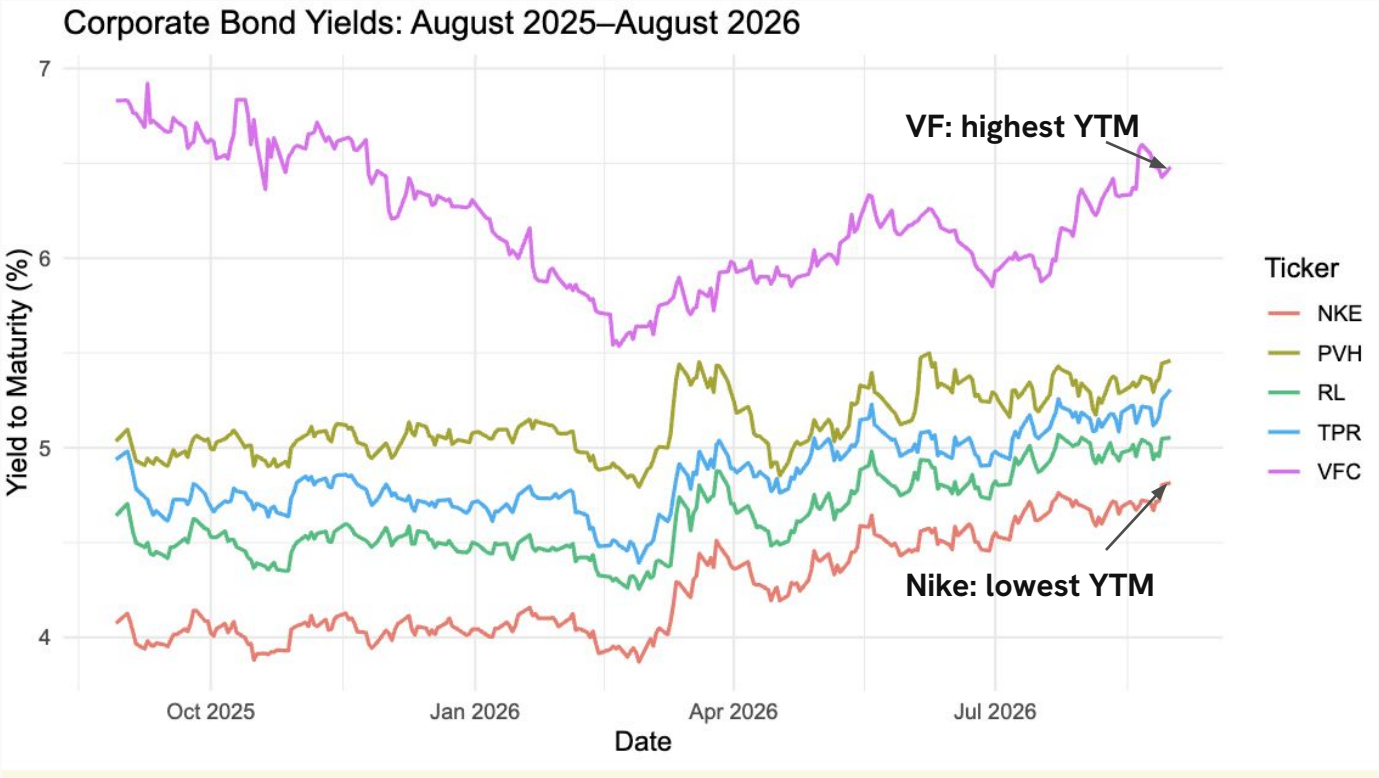
- 100 equals face value.
- Nike and Tapestry stay below 100 throughout the year.
- VF has the widest price range.

Q2(a). Bond Yields (Table 4)

- YTM is in percentage units. A value of 4.8 means 4.8%.

Date	NKE	RL	TPR	PVH	VFC
2025-08-29	4.0746	4.6416	4.9376	5.0342	6.8304
2025-09-02	4.1267	4.7054	4.9821	5.0975	6.8316
2025-09-03	4.0829	4.6231	4.9260	5.0407	6.8046
2025-09-04	4.0235	4.5505	4.8579	4.9862	6.7642
2026-08-26	4.7171	4.9691	5.1355	5.3491	6.4790
2026-08-27	4.7178	4.9543	5.1678	5.3633	6.4676
2026-08-28	4.8024	5.0480	5.2533	5.4442	6.4268
2026-08-31	4.8160	5.0531	5.3074	5.4594	6.4799

Q2(b). Bond Yields (Figure 2)

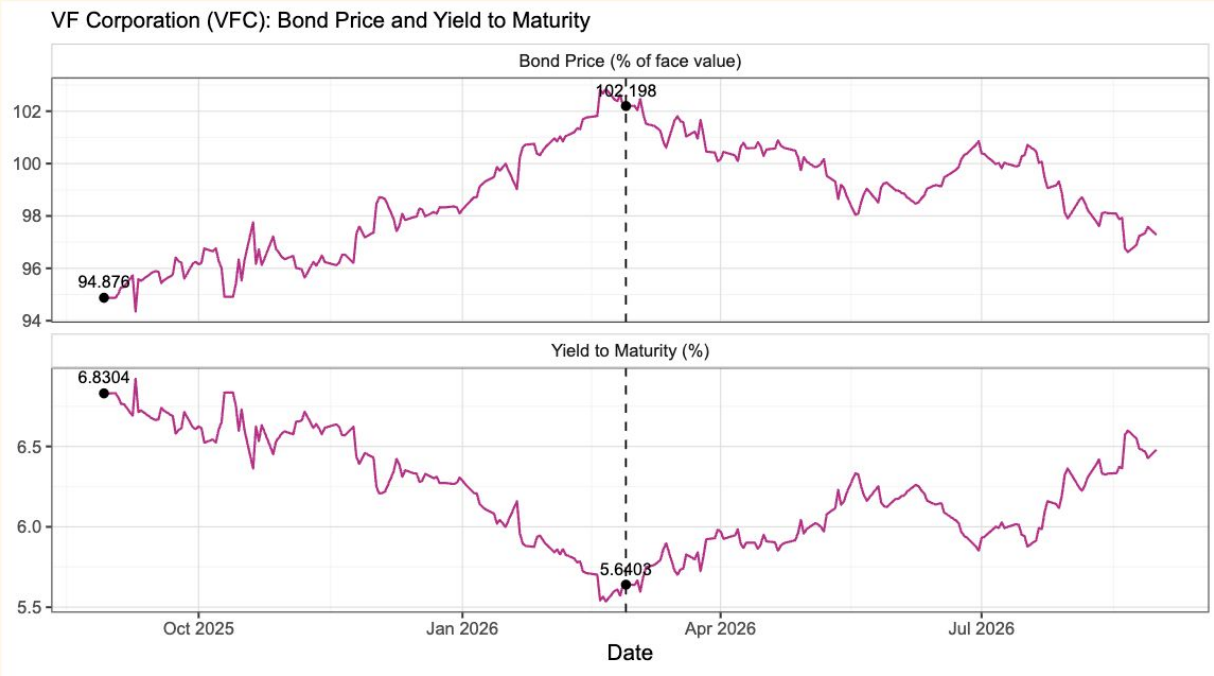


Yield levels

- VF has the highest YTM across the sample.
- Nike has the lowest YTM.
- All five yields rise from early to late March.

Q3(a). Graph Evidence (Figure 2a)

VF Corporation (CUSIP 918204AR9)



VF example

29 August 2025 to

27 February 2026

- Price: 94.876 → 102.198, an increase of 7.72%.
- YTM: 6.8304% → 5.6403%, a fall of 119.01 bps.
- In the shaded interval, price rises while YTM falls.

Q3(a). Economic Reasoning

Economic explanation

- For a given bond, the promised coupons and principal are fixed.
- YTM is the discount rate that equates their present value to the market price.
- If comparable bonds offer higher yields, this bond is less attractive at its old price. Its price falls until its YTM becomes competitive.
- When required returns fall, those cash flows have a higher present value. The price rises and YTM falls.

VF example (Table 5)

VF	29 Aug 2025	27 Feb 2026
Price (% of face)	94.876	102.198
YTM (%)	6.8304	5.6403

Price change = (102.198 / 94.876 − 1) × 100
= +7.72%

YTM change = (5.6403176 − 6.8304134) × 100
= −119.01 bps

Q3(b). Annual Changes (Table 6)

29 August 2025 to 31 August 2026

Company	Price change (%)	Start YTM (%)	End YTM (%)	Change (bps)
Nike	-1.40	4.0746	4.8160	+74.14
Ralph Lauren	-2.30	4.6416	5.0531	+41.15
Tapestry	-0.31	4.9376	5.3074	+36.97
PVH	-1.80	5.0342	5.4594	+42.52
VF Corporation	+2.52	6.8304	6.4799	-35.05

Four bonds ended with lower prices and higher YTMs. VF moved the other way. Price changes exclude coupons.

Calculation

Price change compares the final quote with the initial quote.
100 × (last / first − 1)

YTM change measures the difference in basis points.
100 × (last YTM − first YTM)

Nike example
(93.603 / 94.930 − 1) × 100 = −1.40%
(4.8159582 − 4.0745926) × 100 = +74.14 bps

Q3(b). Common March Move (Table 7)

Bond	2 Mar YTM (%)	26 Mar YTM (%)	Change (bps)
NKE	3.9632	4.5105	+54.73
RL	4.3467	4.8774	+53.07
TPR	4.4863	5.0151	+52.88
PVH	4.8858	5.4343	+54.84
VFC	5.6389	5.8148	+17.59
5-year Treasury	3.62	4.08	+46.00

Measuring the common move

YTM change in basis points
= (26 Mar YTM − 2 Mar YTM) × 100

Nike: (4.51049 − 3.96318) × 100
= +54.73 bps

Treasury: (4.08 − 3.62) × 100
= +46 bps

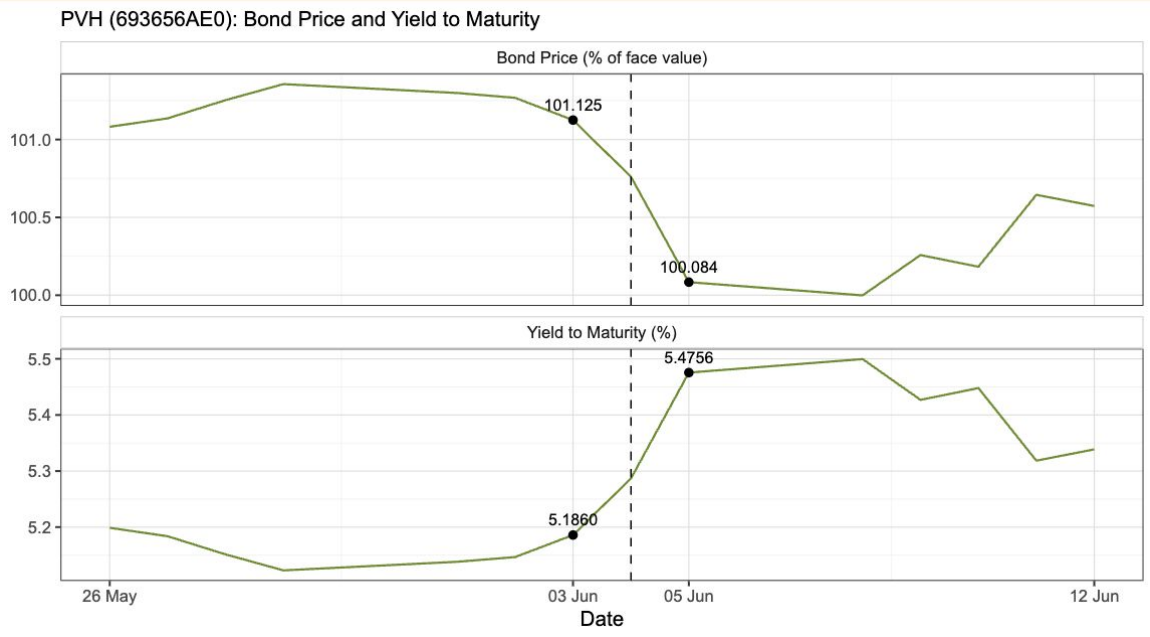
- **All five yields rose and prices fell.**
- The Treasury rise suggests a common market-rate component.

Q3(b). Possible Reasons

- **Oil and inflation:** On 18 March, Powell linked higher oil prices to rising near-term inflation expectations. This can increase the nominal yield investors require.
- **Rate-cut expectations:** Reuters reported on 18 March that traders had pushed expected Fed cuts into 2027. Higher expected future short-term rates can lift longer-term bond yields.
- **Uncertainty:** The March FOMC minutes, released on 8 April, linked higher Treasury term premiums to conflict-related uncertainty. Investors may demand more compensation for holding longer-term bonds.
- **Our interpretation:** Alongside the 46 bps Treasury increase, this supports a common market-rate explanation. Different bond features and issuer risks can still explain differences across the five bonds.

Q3(c). PVH Around the Earnings Release

PVH (CUSIP 693656AE0), Figure 2b



PVH's 3 June results

- Results came after the close on 3 June. The dashed line marks 4 June.
- PVH lowered full-year revenue guidance to approximately flat, citing EMEA demand.
- Weaker expected cash generation may lead investors to require a higher yield.

Q3(c). Interpreting the PVH Move

- **Background:** PVH owns Calvin Klein and Tommy Hilfiger. After the close on 3 June, it lowered its full-year sales forecast from slight growth to roughly unchanged because demand in Europe, the Middle East and Africa was weaker.
- **Why the bond price could fall:** Weaker expected sales may mean less cash to pay interest and repay debt. Investors may then require a higher yield. With the bond's promised payments unchanged, its price would need to fall to offer that higher yield.
- **Why the news was mixed:** PVH kept its adjusted profit-margin and EPS forecasts unchanged, helped by tariff refunds. The sales outlook was weaker, but its profit forecasts were maintained.
- **Observed result:** From 3 to 5 June, the bond price fell from 101.125 to 100.084 (−1.03%), while YTM rose from 5.1860% to 5.4756% (+28.96 bps), which is consistent with the explanation above.

Q4. Pricing Inputs (Table 9)

Input	Value
Bond	Nike, CUSIP 654106AK9
Valuation date	27 March 2026, after the coupon
Maturity	27 March 2030
Selected principal	$2,000
Annual coupon	2.85%
Annual YTM	4.4810017788607%

Calculations

N = 4 years × 2 = 8 periods

C = $2,000 × 0.0285 / 2
= $28.50 every six months

r = (4.4810018 / 100) / 2
= 0.022405009 per period

- The coupon date gives eight full semiannual periods. Divide annual rates by 2 because coupons are paid twice a year.

Q4. Discounted Cash Flows (Table 10)

Period	Cash flow ($)	Discount factor	Present value ($)
1	28.50	0.9780860	27.87545
2	28.50	0.9566522	27.26459
3	28.50	0.9356881	26.66711
4	28.50	0.9151834	26.08273
5	28.50	0.8951280	25.51115
6	28.50	0.8755122	24.95210
7	28.50	0.8563262	24.40530
8	2028.50	0.8375606	1698.99173
Total			1881.75014

Pricing formula

r = 0.022405009 per half-year

$$P = \frac{28.50}{1+r} + \frac{28.50}{(1+r)^2} + \cdots + \frac{28.50}{(1+r)^8} + \frac{2{,}000}{(1+r)^8}$$

Calculated price

$1,881.75

Q4. Calculated vs Observed Price

FactSet Excel export: NIKE

CUSIP 654106AK9 · 27 March 2026

	A	B	C	D	E
1	**Price History: 654106AK9**				
16	**Date**	**Price**	**Change**	**% Change**	**Yield to Maturity**
159	03/25/26	94.474	0.19	0.20	4.368599
160	03/26/26	93.980	-0.49	-0.52	4.510490
161	03/27/26	94.087	0.11	0.11	4.481002
162	03/30/26	94.317	0.23	0.24	4.418373
163	03/31/26	94.449	0.13	0.14	4.378449

Conclusion: The $1,881.75 calculated price is about one cent above the $1,881.74 observed price, so the cash-flow valuation closely matches the market quote.

Price Check

Observed price
94.087 / 100 × $2,000
= $1,881.74

Item	Dollars
Calculated	1,881.75014
Observed	1,881.74000
Difference	0.01014

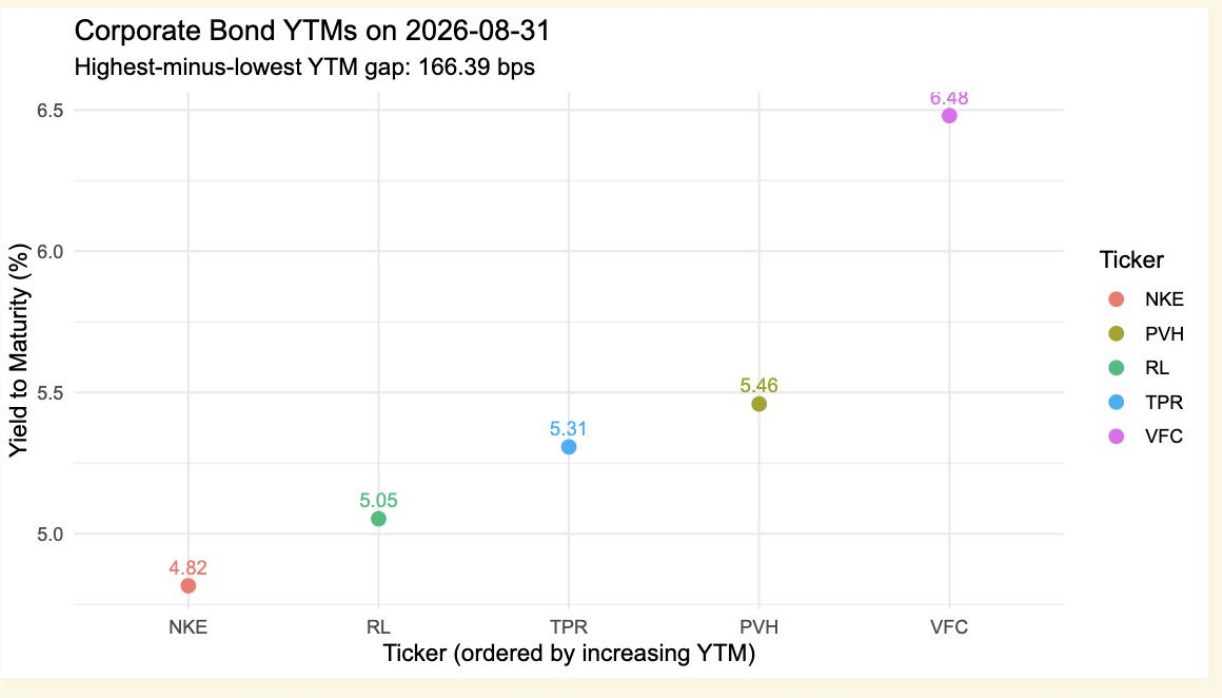
Q5(A). Bond Comparison (Table 12)

As of 31 August 2026. Ordered by increasing YTM.

Company	Coupon Rate (%)	Maturity Date	Years to Maturity	Market price ($)	YTM (%)
Nike	2.85	2030-03-27	3.57	936.03	4.8160
Ralph Lauren	5.00	2032-06-15	5.79	997.13	5.0531
Tapestry	3.05	2032-03-15	5.54	892.73	5.3074
PVH	5.50	2030-06-13	3.78	1,001.12	5.4594
VF Corporation	6.00	2033-10-15	7.12	972.70	6.4799

Dollar price = quote / 100 × $1,000
Nike: 93.603 / 100 × $1,000 = $936.03

Q5(B). Yield Comparison (Figure 3)



Calculation in R

```
last <- subset(
  bond_data,
  date == max(date)
)

gap_bps <- 100 * (
  max(last$ytm) -
  min(last$ytm)
)

print(round(gap_bps, 2))
# [1] 166.39
```

Highest and lowest

VFC: 6.4799%
NKE: 4.8160%
Difference: 166.39 bps

Q5(C)(a). Maturity Comparison (Table 13)

Company	Maturity	YTM on 31 Aug 2026
Nike	27 March 2030	4.815958%
PVH	13 June 2030	5.459417%

Why compare Nike and PVH?

- **Closest maturities:** Across all 10 bond pairs, Nike and PVH have the smallest maturity gap: 78 days.

What the yield gap means

(5.4594166 − 4.8159582) × 100
= 64.35 bps

Q5(C)(b). Yield Differences

- **Credit risk:** FactSet's 1 November snapshots show Nike at A and PVH at BBB. PVH has the lower rating. If investors view PVH as more likely to miss payments, they may require a higher yield. These later snapshots do not establish the ratings on 31 August.
- **Industry:** Both firms depend on consumer spending, but their brands and customers differ. Weaker demand can reduce sales and cash available for debt payments. Their cash-flow risks can therefore differ even within the fashion industry.
- **Coupon and price:** PVH's coupon is 5.50%, versus Nike's 2.85%. YTM reflects both promised payments and the price paid. A higher coupon can be offset by a higher purchase price, so it does not automatically produce a higher YTM.
- **Maturity and other factors:** The bonds mature only 78 days apart, so maturity alone is unlikely to explain the full 64.35 bps gap. A bond that is harder to sell may need to offer a higher yield. The gap may reflect greater perceived risk for PVH, but we cannot attribute it entirely to credit risk.

Appendix

AI Use and Verification

AI contribution

According to the project description, i will draft prompts and pasted them into Copilot. Copilot generated codes and extended the R code in bond_project.R.

Running R

I ran the script in R through VS Code and shared the tables, errors and plots. Selected code and AI replies appear in this appendix.

Verification

The output has 5 companies × 252 dates.
Checks cover missing values, duplicates, date ranges, units and source values.

What I learned

Price is quoted as % of face value.
YTM differences in % units × 100 give bps.
Face value and selected principal differ.

Code Excerpt: Data Loading

```
# Excerpts; helper objects are defined earlier in the script.
raw <- read_excel(path, sheet = "Price History", skip = 15)

bond_data <- pmap_dfr(
  list(file_map$file_name, file_map$ticker, file_map$cusip),
  ~ read_one_bond(..1, ..2, ..3, data_dir = data_dir)
) %>% arrange(date, ticker)

assert_that(all(!is.na(bond_data$date)))
assert_that(all(!is.na(bond_data$price)))
assert_that(all(!is.na(bond_data$ytm)))
assert_that(dup_count == 0)
assert_that(n_companies == 5)
assert_that(all_same_dates)
assert_that(range_ok)
```

Code Excerpt: Q4 Pricing

```
# NKE, 2026-03-27; principal_amount = 2000, n_periods = 8
price_pct <- nke_obs$price[[1]]
ytm_pct <- nke_obs$ytm[[1]]
coupon_payment <- principal_amount * (coupon_rate_pct / 100) / 2
semiannual_yield <- (ytm_pct / 100) / 2
period_index <- seq_len(n_periods)
cash_flow <- rep(coupon_payment, n_periods)
cash_flow[n_periods] <- cash_flow[n_periods] + principal_amount
discount_factor <- 1 / ((1 + semiannual_yield)^period_index)
present_value <- cash_flow * discount_factor
calculated_price <- sum(present_value)
observed_price <- price_pct / 100 * principal_amount
price_difference <- calculated_price - observed_price

# Calculated: 1881.75014; observed: 1881.74000
# Difference: 0.01014; Excel SUM() and PV() give the same price.
```

AI Record: Loading and Validation

Prompt excerpts

- “The worksheet is called "Price History".”
- “Column headers are on row 16, so use skip = 15.”
- “Use assert_that() to check for missing prices or yields, duplicate company-date pairs, five companies, matching dates, and the expected range 2025-08-29 through 2026-08-31.”
- “Only complete loading and validation for now.”

Copilot reply excerpts

- “Done. I created bond_project.R with simple, commented code that:”
- “Reads all 5 Excel files from data/ (sheet "Price History", skip = 15)”
- “Combines and sorts by date, ticker”
- “Validates using assertthat::assert_that()”

R check: 252 rows per company; no missing values or duplicate company-date pairs. All validations passed.

AI Record: Descriptive Statistics

Request summary

- Request summary: Append start-to-end price and YTM changes using bond_data. Identify each company's three largest absolute daily YTM moves. Keep the earlier analysis unchanged.
- What I learned: Sort observations by ticker and date before using lag(). A daily change uses the previous available observation, which may be several calendar days earlier.

Copilot reply excerpts

- “Done. I appended a short Question 3 preparation section to bond_project.R, with existing code unchanged.”
- “Computes the 3 largest absolute daily YTM moves per ticker”
- Verification: Results were checked against original Excel rows. The 3–5 June event comparison spans two trading intervals and is different from one daily move.

AI Record: Plotting Error and Fix

R error shared in chat:
unused argument (weight = if (bold) "bold" else "normal")

```
install.packages("systemfonts", repos = "https://cloud.r-project.org")
packageVersion("systemfonts")
"weight" %in% names(formals(systemfonts::font_info))
# Returned: version 1.3.2; TRUE
```

- **AI guidance (summary):** Update systemfonts, restart R, and check that font_info() accepts weight.
- **Verification:** The package check returned 1.3.2 and supports_weight = TRUE. The price and yield plots then rendered.
- **What I learned:** A graphics-package error does not mean the bond data are wrong.

AI Record: Correcting the Price Basis

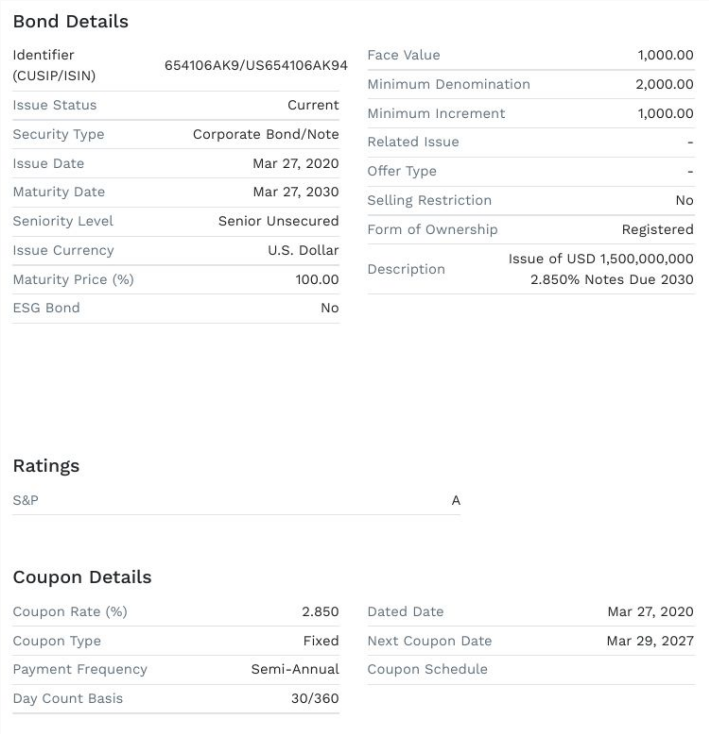
- **Source check:** FactSet lists $1,000 face value for every bond. Minimum denominations are $2,000 for NKE, RL, TPR and PVH, and $1,000 for VFC.
- **Correction to AI output:** I kept Q4's selected $2,000 principal, but changed Q5 to report the dollar price per $1,000 face value.
- **What I learned:** Face value, minimum denomination and selected principal describe different things. Each calculation must label the amount being valued.

```
# Q5 excerpt from mutate():
dollar_price_per_face_value = price / 100 * face_value_usd

# R output on 31 August 2026, in USD:
# NKE 936.03; RL 997.13; TPR 892.73; PVH 1001.12; VFC 972.70
```

Nike Bond Information (Figure A1)

FactSet snapshot: 1 November 2026



Bond Details

Identifier (CUSIP/ISIN)	654106AK9/US654106AK94
Issue Status	Current
Security Type	Corporate Bond/Note
Issue Date	Mar 27, 2020
Maturity Date	Mar 27, 2030
Seniority Level	Senior Unsecured
Issue Currency	U.S. Dollar
Maturity Price (%)	100.00
ESG Bond	No

Face Value	1,000.00
Minimum Denomination	2,000.00
Minimum Increment	1,000.00
Related Issue	-
Offer Type	-
Selling Restriction	No
Form of Ownership	Registered
Description	Issue of USD 1,500,000,000 2.850% Notes Due 2030

Ratings

S&P	A

Coupon Details

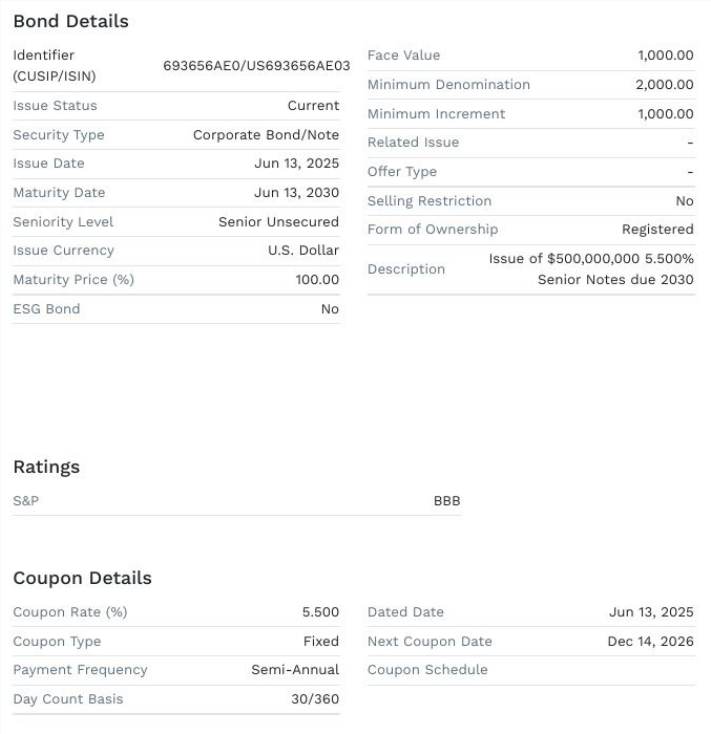
Coupon Rate (%)	2.850
Coupon Type	Fixed
Payment Frequency	Semi-Annual
Day Count Basis	30/360

Dated Date	Mar 27, 2020
Next Coupon Date	Mar 29, 2027
Coupon Schedule	

- CUSIP: 654106AK9
- Coupon: 2.85%, semiannual
- Maturity: 2030-03-27
- Face value: $1,000
- Minimum denomination: $2,000
- S&P snapshot rating: A

PVH Bond Information (Figure A2)

FactSet snapshot: 1 November 2026



Bond Details

Identifier (CUSIP/ISIN)	693656AE0/US693656AE03	Face Value	1,000.00
Issue Status	Current	Minimum Denomination	2,000.00
Security Type	Corporate Bond/Note	Minimum Increment	1,000.00
Issue Date	Jun 13, 2025	Related Issue	-
Maturity Date	Jun 13, 2030	Offer Type	-
Seniority Level	Senior Unsecured	Selling Restriction	No
Issue Currency	U.S. Dollar	Form of Ownership	Registered
Maturity Price (%)	100.00	Description	Issue of $500,000,000 5.500% Senior Notes due 2030
ESG Bond	No		

Ratings

S&P	BBB

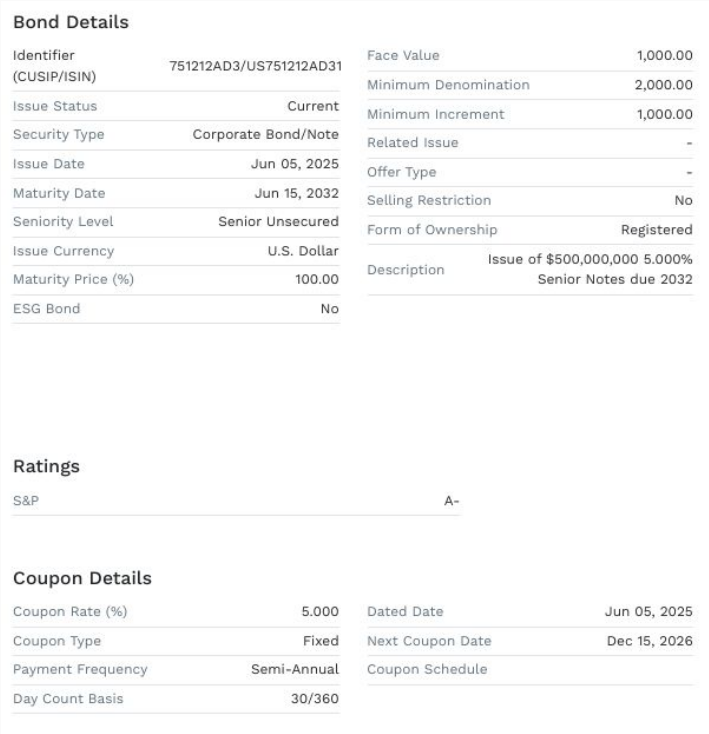
Coupon Details

Coupon Rate (%)	5.500	Dated Date	Jun 13, 2025
Coupon Type	Fixed	Next Coupon Date	Dec 14, 2026
Payment Frequency	Semi-Annual	Coupon Schedule	
Day Count Basis	30/360		

- CUSIP: 693656AE0
- Coupon: 5.50%, semiannual
- Maturity: 2030-06-13
- Face value: $1,000
- Minimum denomination: $2,000
- S&P snapshot rating: BBB

Ralph Lauren Bond Information (Figure A3)

FactSet snapshot: 1 November 2026



Bond Details

Identifier (CUSIP/ISIN)	751212AD3/US751212AD31	Face Value	1,000.00
Issue Status	Current	Minimum Denomination	2,000.00
Security Type	Corporate Bond/Note	Minimum Increment	1,000.00
Issue Date	Jun 05, 2025	Related Issue	-
Maturity Date	Jun 15, 2032	Offer Type	-
Seniority Level	Senior Unsecured	Selling Restriction	No
Issue Currency	U.S. Dollar	Form of Ownership	Registered
Maturity Price (%)	100.00	Description	Issue of $500,000,000 5.000% Senior Notes due 2032
ESG Bond	No		

Ratings

S&P	A-

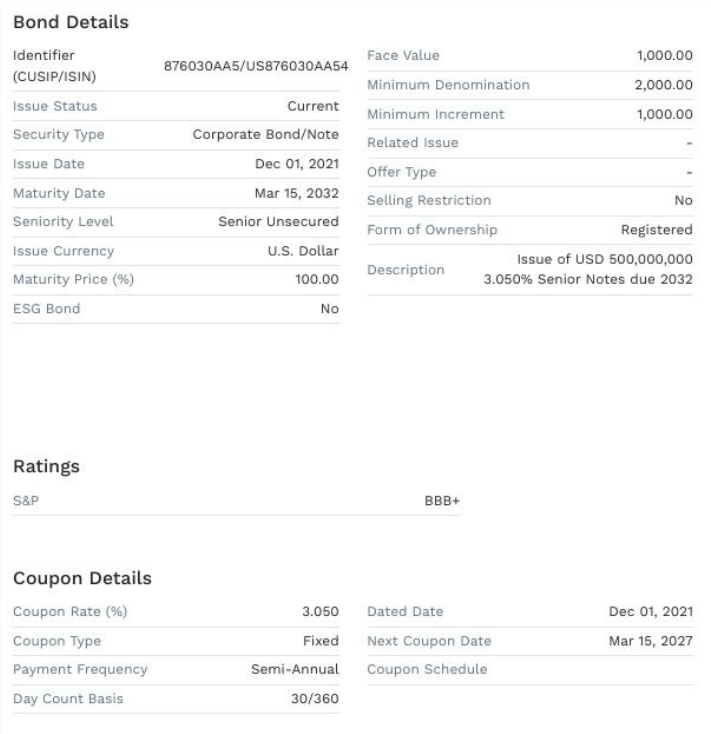
Coupon Details

Coupon Rate (%)	5.000	Dated Date	Jun 05, 2025
Coupon Type	Fixed	Next Coupon Date	Dec 15, 2026
Payment Frequency	Semi-Annual	Coupon Schedule	
Day Count Basis	30/360		

- CUSIP: 751212AD3
- Coupon: 5.00%, semiannual
- Maturity: 2032-06-15
- Face value: $1,000
- Minimum denomination: $2,000
- S&P snapshot rating: A-

Tapestry Bond Information (Figure A4)

FactSet snapshot: 1 November 2026



Bond Details

Identifier (CUSIP/ISIN)	876030AA5/US876030AA54	Face Value	1,000.00
Issue Status	Current	Minimum Denomination	2,000.00
Security Type	Corporate Bond/Note	Minimum Increment	1,000.00
Issue Date	Dec 01, 2021	Related Issue	-
Maturity Date	Mar 15, 2032	Offer Type	-
Seniority Level	Senior Unsecured	Selling Restriction	No
Issue Currency	U.S. Dollar	Form of Ownership	Registered
Maturity Price (%)	100.00	Description	Issue of USD 500,000,000 3.050% Senior Notes due 2032
ESG Bond	No		

Ratings

S&P	BBB+

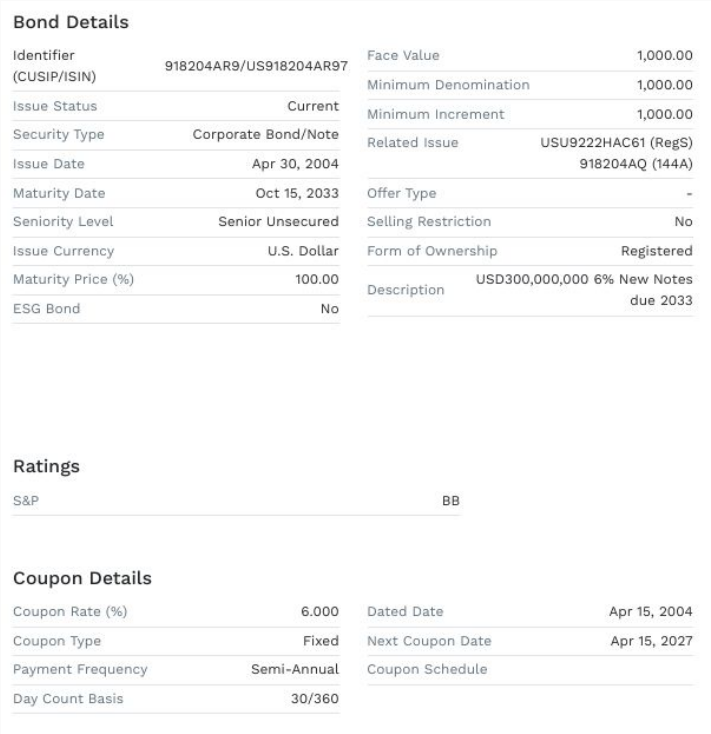
Coupon Details

Coupon Rate (%)	3.050	Dated Date	Dec 01, 2021
Coupon Type	Fixed	Next Coupon Date	Mar 15, 2027
Payment Frequency	Semi-Annual	Coupon Schedule	
Day Count Basis	30/360		

- CUSIP: 876030AA5
- Coupon: 3.05%, semiannual
- Maturity: 2032-03-15
- Face value: $1,000
- Minimum denomination: $2,000
- S&P snapshot rating: BBB+

VF Corporation Bond Information (Figure A5)

FactSet snapshot: 1 November 2026



Bond Details

Identifier (CUSIP/ISIN)	918204AR9/US918204AR97	Face Value	1,000.00
Issue Status	Current	Minimum Denomination	1,000.00
Security Type	Corporate Bond/Note	Minimum Increment	1,000.00
Issue Date	Apr 30, 2004	Related Issue	USU9222HAC61 (RegS) 918204AQ (144A)
Maturity Date	Oct 15, 2033	Offer Type	-
Seniority Level	Senior Unsecured	Selling Restriction	No
Issue Currency	U.S. Dollar	Form of Ownership	Registered
Maturity Price (%)	100.00	Description	USD300,000,000 6% New Notes due 2033
ESG Bond	No		

Ratings

S&P	BB

Coupon Details

Coupon Rate (%)	6.000	Dated Date	Apr 15, 2004
Coupon Type	Fixed	Next Coupon Date	Apr 15, 2027
Payment Frequency	Semi-Annual	Coupon Schedule	
Day Count Basis	30/360		

- CUSIP: 918204AR9
- Coupon: 6.00%, semiannual
- Maturity: 2033-10-15
- Face value: $1,000
- Minimum denomination: $1,000
- S&P snapshot rating: BB

Reference Links and Data Files

Federal Reserve press conference, 18 March 2026

U.S. Treasury daily par yields, March 2026

PVH first-quarter results, 3 June 2026

Course assignment and example code